US012414841B2

(12) United States Patent
Khardekar et al.

(10) Patent No.: US 12,414,841 B2

(45) Date of Patent: *Sep. 16, 2025

(54) **METHODS AND SYSTEMS FOR INTERPROXIMAL REDUCTION PLANNING**

(71) Applicant: **Align Technology, Inc.**, San Jose, CA (US)

(72) Inventors: **Rahul Khardekar**, Sunnyvale, CA (US); **Artem Borovinskih**, San Jose, CA (US); **Rene M. Sterental**, Palo Alto, CA (US); **Giovanny Garro Acevedo**, La Union (CR); **Jason Ramos**, San Jose (CR); **Mitra Derakhshan**, Herndon, VA (US); **Sergey Kurchatov**, Troitsk (RU); **Igor Kvasov**, San Jose, CA (US)

(73) Assignee: **Align Technology, Inc.**, San Jose, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/146,348**

(22) Filed: **Dec. 23, 2022**

(65) **Prior Publication Data**

US 2023/0125417 A1 Apr. 27, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/776,369, filed on Jan. 29, 2020, now Pat. No. 11,534,265, which is a (Continued)

(51) **Int. Cl.**

A61C 7/00	(2006.01)
G06F 30/00	(2020.01)

(52) **U.S. Cl.**

CPC ***A61C 7/002*** (2013.01); ***G06F 30/00*** (2020.01)

(58) **Field of Classification Search**

CPC A61C 7/002; G06F 30/00

(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,820,368 A	10/1998	Wolk		
6,183,248 B1 *	2/2001	Chishti	....	A61C 7/08 433/24

(Continued)

OTHER PUBLICATIONS

Livas et al. ("Enamel Reduction Techniques in Orthodontics: A Literature Review", The Open Dentistry Journal, 2013, 7, 146-151) (Year: 2013).*

(Continued)

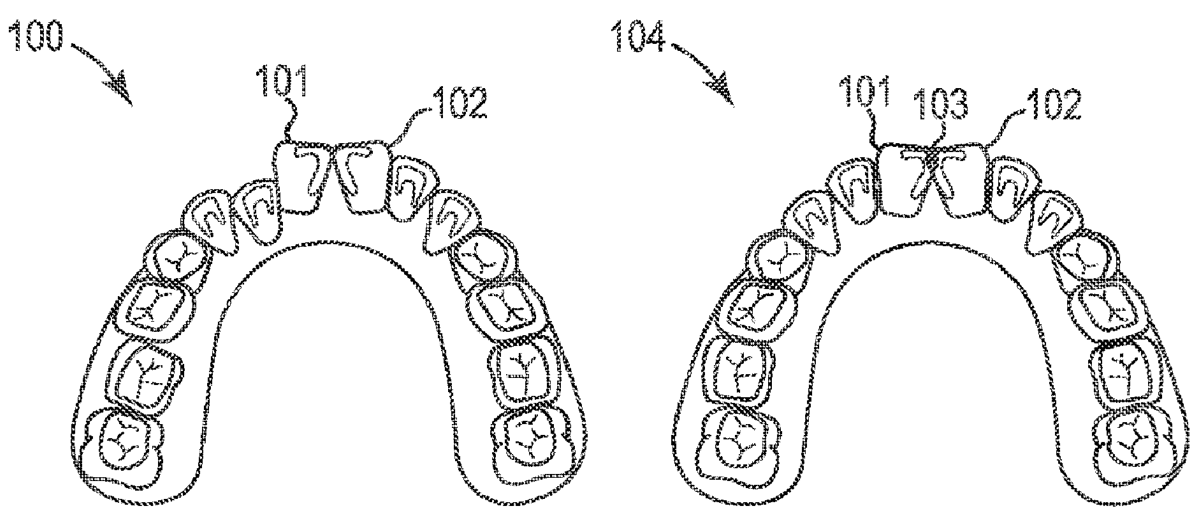
Primary Examiner — Iftekhar A Khan

(74) *Attorney, Agent, or Firm* — Shay Glenn LLP

(57) **ABSTRACT**

Methods and systems for IPR planning. One method includes identifying an overlap between a first tooth and a second tooth in a patient's dentition at a target position of a treatment plan; identifying first and second ridge endpoint reference lines on each of the first tooth and the second tooth; calculating a distance difference between the first and second ridge endpoint reference lines; determining that the distance difference is outside a threshold difference; revising the treatment plan by adding one or more stages to the treatment plan where the distance difference between the first and second ridge endpoint reference lines is within the threshold difference; and prescribing IPR on one or both of the first and second teeth at the one or more additional stages of the treatment plan.

20 Claims, 5 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/253,148, filed on Aug. 31, 2016, now Pat. No. 10,595,965, which is a continuation of application No. 13/410,153, filed on Mar. 1, 2012, now Pat. No. 9,433,476.

(58) **Field of Classification Search**
USPC 703/2
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,309,215 B1 10/2001 Phan et al.
6,386,864 B1 5/2002 Kuo
6,406,292 B1 * 6/2002 Chishti G16H 50/50 433/213
6,454,565 B2 9/2002 Phan et al.
6,471,511 B1 10/2002 Chishti et al.
6,524,101 B1 2/2003 Phan et al.
6,572,372 B1 6/2003 Phan et al.
6,607,382 B1 8/2003 Kuo et al.
6,705,863 B2 3/2004 Phan et al.
6,783,604 B2 8/2004 Tricca
6,790,035 B2 9/2004 Tricca et al.
6,814,574 B2 11/2004 Abolfathi et al.
6,830,450 B2 12/2004 Knopp et al.
6,947,038 B1 9/2005 Anh et al.
7,074,039 B2 7/2006 Kopelman et al.
7,104,792 B2 9/2006 Taub et al.
7,121,825 B2 10/2006 Chishti et al.
7,160,107 B2 1/2007 Kopelman et al.
7,192,273 B2 3/2007 McSurdy, Jr.
7,347,688 B2 3/2008 Kopelman et al.
7,354,270 B2 * 4/2008 Abolfathi A61C 7/36 606/105
7,448,514 B2 11/2008 Wen
7,481,121 B1 1/2009 Cao
7,543,511 B2 6/2009 Kimura et al.
7,553,157 B2 6/2009 Abolfathi et al.
7,578,673 B2 * 8/2009 Wen A61C 7/146 433/24
7,585,172 B2 * 9/2009 Rubbert A61C 9/006 433/24
7,600,999 B2 10/2009 Knopp
7,689,398 B2 * 3/2010 Cheng A61C 7/00 703/2
7,746,339 B2 * 6/2010 Matov A61C 19/04 433/213
7,766,658 B2 8/2010 Tricca et al.
7,771,195 B2 8/2010 Knopp et al.
7,854,609 B2 12/2010 Chen et al.
7,871,269 B2 1/2011 Wu et al.
7,878,801 B2 2/2011 Abolfathi et al.
7,878,805 B2 2/2011 Moss et al.
7,883,334 B2 2/2011 Li et al.
7,914,283 B2 3/2011 Kuo
7,947,508 B2 5/2011 Tricca et al.
8,152,518 B2 4/2012 Kuo
8,172,569 B2 5/2012 Matty et al.
8,235,715 B2 8/2012 Kuo
8,292,617 B2 10/2012 Brandt et al.
8,337,199 B2 12/2012 Wen
8,401,686 B2 3/2013 Moss et al.
8,439,672 B2 * 5/2013 Matov A61C 7/20 433/24
8,517,726 B2 8/2013 Kakavand et al.
8,562,337 B2 10/2013 Kuo et al.
8,684,729 B2 4/2014 Wen
8,708,697 B2 4/2014 Li et al.
8,758,009 B2 6/2014 Chen et al.
8,771,149 B2 7/2014 Rahman et al.
8,899,976 B2 12/2014 Chen et al.
8,899,977 B2 * 12/2014 Cao A61C 7/08 433/24
8,936,463 B2 1/2015 Mason et al.
8,936,464 B2 1/2015 Kopelman
9,119,691 B2 9/2015 Namiranian et al.
9,161,823 B2 10/2015 Morton et al.
9,241,774 B2 1/2016 Li et al.
9,326,831 B2 5/2016 Cheang
9,433,476 B2 * 9/2016 Khardekar G06F 30/00
9,610,141 B2 4/2017 Kopelman et al.
9,655,691 B2 5/2017 Li et al.
9,675,427 B2 6/2017 Kopelman
9,700,385 B2 7/2017 Webber
9,744,001 B2 8/2017 Choi et al.
9,844,424 B2 12/2017 Wu et al.
10,045,835 B2 8/2018 Boronkay et al.
10,111,730 B2 10/2018 Webber et al.
10,150,244 B2 12/2018 Sato et al.
10,213,277 B2 2/2019 Webber et al.
10,299,894 B2 5/2019 Tanugula et al.
10,363,116 B2 7/2019 Boronkay
10,383,705 B2 8/2019 Shanjani et al.
D865,180 S 10/2019 Bauer et al.
10,449,016 B2 10/2019 Kimura et al.
10,463,452 B2 11/2019 Matov et al.
10,470,847 B2 11/2019 Shanjani et al.
10,492,888 B2 12/2019 Chen et al.
10,517,701 B2 12/2019 Boronkay
10,537,406 B2 1/2020 Wu et al.
10,537,463 B2 1/2020 Kopelman
10,548,700 B2 2/2020 Fernie
10,588,776 B2 3/2020 Cam et al.
10,595,965 B2 * 3/2020 Khardekar G06F 30/00
10,613,515 B2 4/2020 Cramer et al.
10,639,134 B2 5/2020 Shanjani et al.
10,743,964 B2 8/2020 Wu et al.
10,781,274 B2 9/2020 Liska et al.
10,792,127 B2 * 10/2020 Kopelman A61C 7/002
10,813,720 B2 10/2020 Grove et al.
10,874,483 B2 12/2020 Boronkay
10,881,487 B2 1/2021 Cam et al.
10,912,629 B2 2/2021 Tanugula et al.
10,959,810 B2 3/2021 Li et al.
10,993,783 B2 5/2021 Wu et al.
11,026,768 B2 6/2021 Moss et al.
11,026,831 B2 6/2021 Kuo
11,045,283 B2 6/2021 Riley et al.
11,103,330 B2 8/2021 Webber et al.
11,123,156 B2 9/2021 Cam et al.
11,166,788 B2 11/2021 Webber
11,174,338 B2 11/2021 Liska et al.
11,219,506 B2 1/2022 Shanjani et al.
11,259,896 B2 3/2022 Matov et al.
11,273,011 B2 3/2022 Shanjani et al.
11,278,375 B2 3/2022 Wang et al.
11,318,667 B2 5/2022 Mojdeh et al.
11,331,166 B2 5/2022 Morton et al.
11,344,385 B2 5/2022 Morton et al.
11,376,101 B2 7/2022 Sato et al.
11,419,702 B2 8/2022 Sato et al.
11,471,253 B2 10/2022 Venkatasanthanam et al.
11,497,586 B2 11/2022 Kopelman
11,504,214 B2 11/2022 Wu et al.
11,523,881 B2 12/2022 Wang et al.
11,534,265 B2 * 12/2022 Khardekar A61C 7/002
11,534,268 B2 12/2022 Li et al.
11,534,974 B2 12/2022 O'Leary et al.
11,554,000 B2 1/2023 Webber
11,564,777 B2 1/2023 Kopelman et al.
11,571,278 B2 2/2023 Kopelman et al.
11,571,279 B2 2/2023 Wang et al.
11,576,750 B2 2/2023 Kopelman et al.
11,576,752 B2 2/2023 Morton et al.
2002/0064746 A1 * 5/2002 Muhammad A61C 7/00 433/24
2002/0192617 A1 12/2002 Phan et al.
2004/0029068 A1 * 2/2004 Sachdeva G16H 50/50 433/24
2004/0137400 A1 * 7/2004 Chishti A61C 7/00 433/24
2004/0166462 A1 8/2004 Phan et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2004/0166463	A1*	8/2004	Wen	A61C 7/146 433/24
2004/0197727	A1*	10/2004	Sachdeva	A61C 7/00 433/24
2005/0014105	A1	1/2005	Abolfathi et al.	
2005/0186524	A1	8/2005	Abolfathi et al.	
2005/0244768	A1	11/2005	Taub et al.	
2005/0244791	A1*	11/2005	Davis	A61C 7/00 433/213
2005/0271996	A1*	12/2005	Sporbert	A61C 7/00 433/24
2006/0019218	A1	1/2006	Kuo	
2006/0078841	A1	4/2006	DeSimone et al.	
2006/0115782	A1	6/2006	Li et al.	
2006/0115785	A1	6/2006	Li et al.	
2006/0194163	A1*	8/2006	Tricca	A61C 7/00 433/24
2006/0199142	A1	9/2006	Liu et al.	
2006/0234179	A1	10/2006	Wen et al.	
2006/0275736	A1*	12/2006	Wen	A61C 9/00 433/213
2008/0057461	A1*	3/2008	Cheng	A61C 7/00 433/24
2008/0118882	A1	5/2008	Su	
2008/0160473	A1	7/2008	Li et al.	
2008/0286716	A1	11/2008	Sherwood	
2008/0286717	A1	11/2008	Sherwood	
2009/0029310	A1*	1/2009	Pumphrey	A61C 7/08 433/24
2009/0280450	A1	11/2009	Kuo	
2009/0286196	A1*	11/2009	Wen	A61C 7/002 433/24
2010/0055635	A1	3/2010	Kakavand	
2010/0129763	A1	5/2010	Kuo	
2010/0280798	A1*	11/2010	Pattijn	A61C 7/002 703/1
2011/0269092	A1	11/2011	Kuo et al.	
2012/0150494	A1*	6/2012	Anderson	A61C 7/08 703/1
2013/0089828	A1*	4/2013	Borovinskih	A61C 19/05 433/24
2013/0204583	A1*	8/2013	Matov	A61C 7/08 703/1
2013/0204599	A1*	8/2013	Matov	G16H 50/50 703/11
2013/0209952	A1*	8/2013	Kuo	A61C 7/12 433/10
2013/0317800	A1*	11/2013	Wu	G06T 19/20 703/11
2013/0325431	A1*	12/2013	See	A61C 7/002 703/11
2014/0067334	A1	3/2014	Kuo	
2014/0379356	A1*	12/2014	Sachdeva	A61C 7/002 705/2
2015/0216627	A1*	8/2015	Kopelman	A61C 7/08 433/24
2015/0366637	A1*	12/2015	Kopelman	A61C 7/08 83/13
2015/0366638	A1	12/2015	Kopelman et al.	
2016/0175068	A1*	6/2016	Cai	A61C 7/002 700/98
2016/0193014	A1*	7/2016	Morton	A61C 7/06 433/24
2016/0310236	A1*	10/2016	Kopelman	G06F 30/00
2016/0367339	A1*	12/2016	Khardekar	G06F 30/00
2017/0007359	A1	1/2017	Kopelman et al.	
2017/0007361	A1	1/2017	Boronkay et al.	
2017/0007365	A1*	1/2017	Kopelman	A61C 7/002
2017/0007386	A1*	1/2017	Mason	G06Q 30/0643
2017/0008333	A1*	1/2017	Mason	B44C 1/228
2017/0100207	A1*	4/2017	Wen	A61C 9/004
2017/0135793	A1	5/2017	Webber et al.	
2017/0165032	A1	6/2017	Webber et al.	
2018/0360567	A1	12/2018	Xue et al.	
2019/0000592	A1	1/2019	Cam et al.	
2019/0000593	A1	1/2019	Cam et al.	
2019/0046297	A1	2/2019	Kopelman et al.	
2019/0099129	A1	4/2019	Kopelman et al.	
2019/0125497	A1	5/2019	Derakhshan et al.	
2019/0175303	A1*	6/2019	Akopov	A61C 7/002
2019/0262101	A1	8/2019	Shanjani et al.	
2019/0298494	A1	10/2019	Webber et al.	
2020/0000553	A1	1/2020	Makarenkova et al.	
2020/0100866	A1	4/2020	Medvinskaya et al.	
2020/0155276	A1	5/2020	Cam et al.	
2020/0188062	A1	6/2020	Kopelman et al.	
2020/0214598	A1	7/2020	Li et al.	
2020/0214801	A1	7/2020	Wang et al.	
2020/0268484	A1*	8/2020	Khardekar	G06F 30/00
2020/0390523	A1	12/2020	Sato et al.	
2021/0147672	A1	5/2021	Cole et al.	
2023/0125417	A1*	4/2023	Khardekar	G06F 30/00 703/2

OTHER PUBLICATIONS

Clément Frindel ("Clear thinking about interproximal stripping", J Dentofacial Anom Orthod 2010;13:187-199) (Year: 2010).*

* cited by examiner

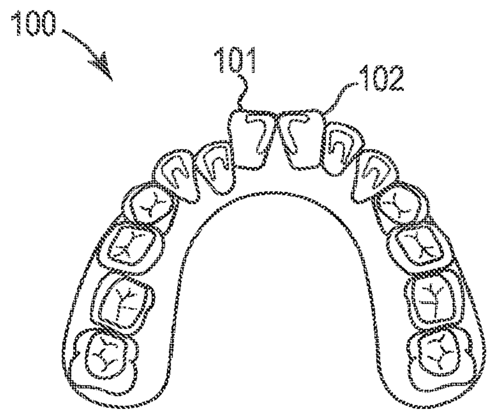


Fig. 1A

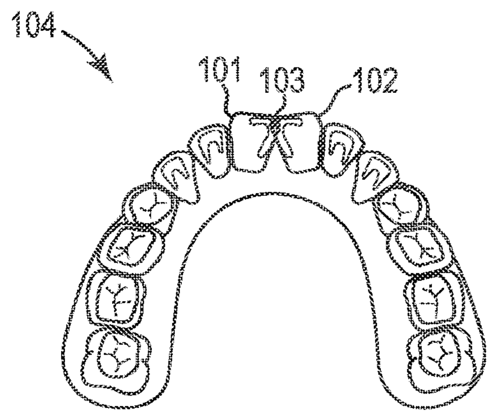


Fig. 1B

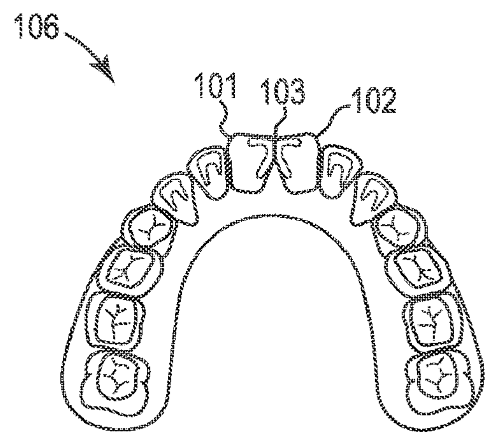


Fig. 1C

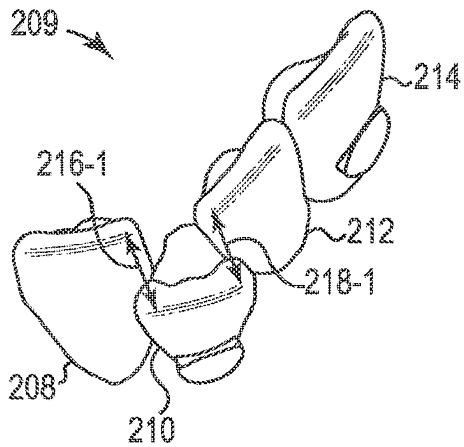


Fig. 2A

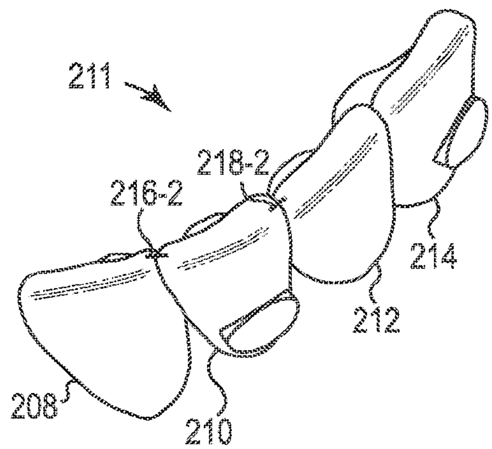


Fig. 2B

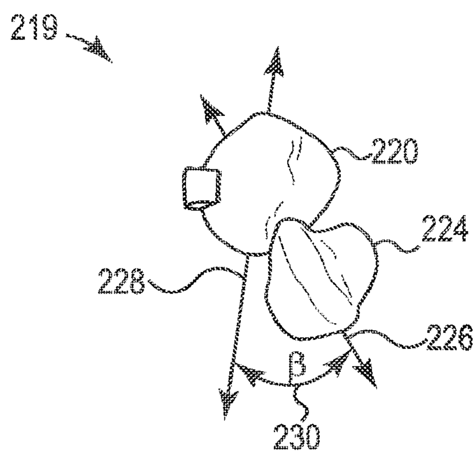


Fig. 2C

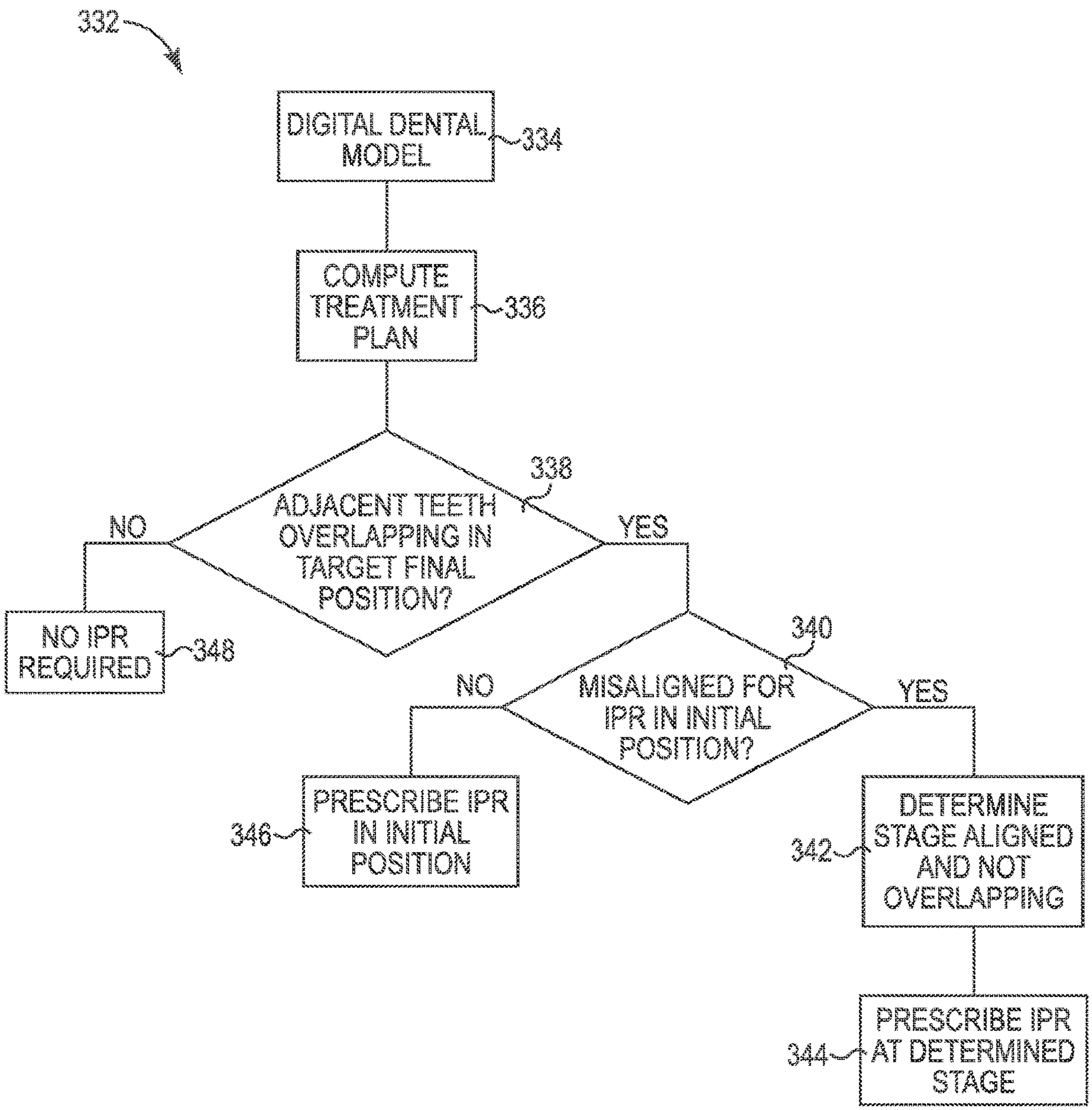

332
DIGITAL DENTAL MODEL
334
COMPUTE TREATMENT PLAN
336
ADJACENT TEETH OVERLAPPING IN TARGET FINAL POSITION?
338
NO
YES
NO IPR REQUIRED
348
MISALIGNED FOR IPR IN INITIAL POSITION?
340
NO
YES
PRESCRIBE IPR IN INITIAL POSITION
346
DETERMINE STAGE ALIGNED AND NOT OVERLAPPING
342
PRESCRIBE IPR AT DETERMINED STAGE
344

Fig. 3

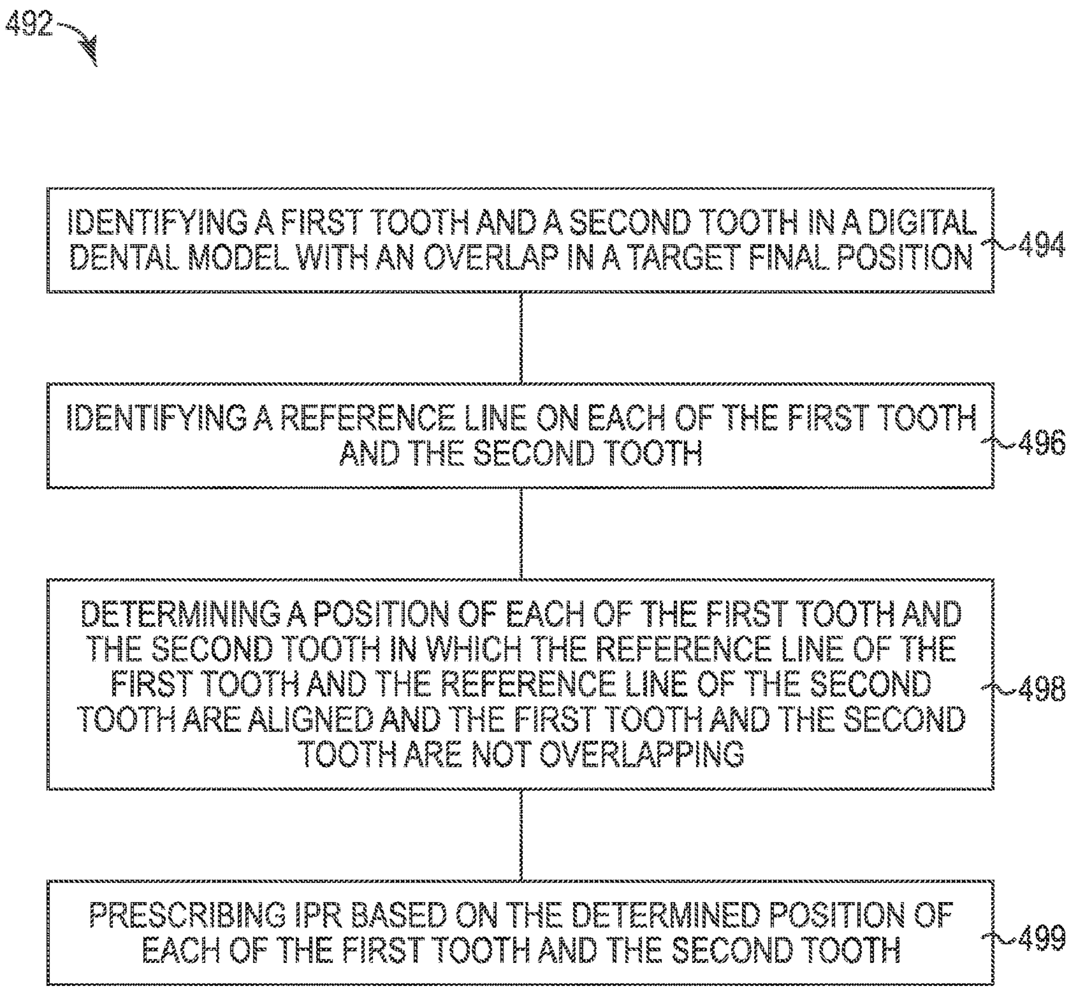

492
IDENTIFYING A FIRST TOOTH AND A SECOND TOOTH IN A DIGITAL DENTAL MODEL WITH AN OVERLAP IN A TARGET FINAL POSITION
494
IDENTIFYING A REFERENCE LINE ON EACH OF THE FIRST TOOTH AND THE SECOND TOOTH
496
DETERMINING A POSITION OF EACH OF THE FIRST TOOTH AND THE SECOND TOOTH IN WHICH THE REFERENCE LINE OF THE FIRST TOOTH AND THE REFERENCE LINE OF THE SECOND TOOTH ARE ALIGNED AND THE FIRST TOOTH AND THE SECOND TOOTH ARE NOT OVERLAPPING
498
PRESCRIBING IPR BASED ON THE DETERMINED POSITION OF EACH OF THE FIRST TOOTH AND THE SECOND TOOTH
499

Fig. 4

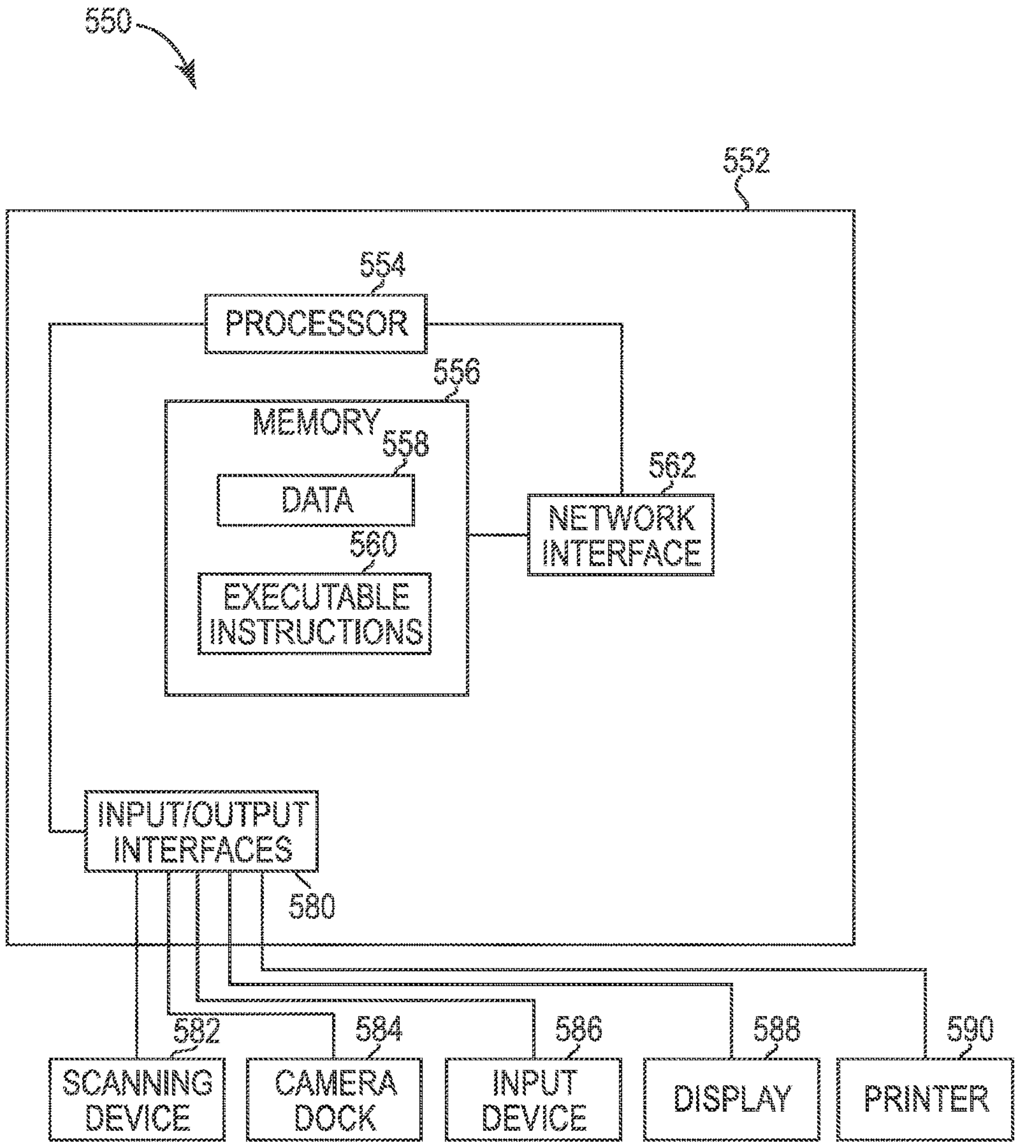

550
552
554
PROCESSOR
556
MEMORY
558
DATA
560
EXECUTABLE INSTRUCTIONS
562
NETWORK INTERFACE
INPUT/OUTPUT INTERFACES
580
582
SCANNING DEVICE
584
CAMERA DOCK
586
INPUT DEVICE
588
DISPLAY
590
PRINTER

Fig. 5

METHODS AND SYSTEMS FOR INTERPROXIMAL REDUCTION PLANNING

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 16/776,369 filed on Jan. 29, 2020, now U.S. Pat. No. 11,534,265, which is a continuation of U.S. patent application Ser. No. 15/253,148, filed on Aug. 31, 2016, now U.S. Pat. No. 10,595,965, which is a continuation of U.S. patent application Ser. No. 13/410,153, filed on Mar. 1, 2012, now U.S. Pat. No. 9,433,476, each of which is herein incorporated by reference in its entirety.

BACKGROUND

The present disclosure is related generally to the field of dental treatment. More particularly, the present disclosure relates to interproximal reduction planning.

Many dental treatments involve repositioning misaligned teeth and changing bite configurations for improved cosmetic appearance and dental function. Orthodontic repositioning can be accomplished, for example, through a dental process that uses one or more removable positioning appliances for realigning teeth.

Such appliances may utilize a shell of material having resilient properties, referred to as an "aligner" that generally conforms to a patient's teeth but is slightly out of alignment with the initial tooth configuration. Placement of an appliance over the teeth can provide controlled forces in specific locations to gradually move the teeth into a new configuration. Repetition of this process with successive appliances in progressive configurations can move the teeth through a series of intermediate arrangements to a final desired arrangement.

Repositioning a patient's teeth may result in residual crowding of adjacent teeth due to insufficient space within the patient's mouth. This residual crowding can impede complete tooth alignment. In some situations it may be possible to remove a small portion of a tooth, or portions of two adjacent teeth, in order to make the teeth fit within the space available. The removal of material causing the overlap of the crowded teeth must be treated by the treatment professional by removing material from the surface of one or more teeth in a process called interproximal reduction (IPR). During an IPR procedure, a small amount of enamel thickness on the surface of the teeth is removed to reduce the mesial-distal width and space requirements for the tooth.

One problem experienced during dental treatment is the determination by the treatment professional of whether an IPR procedure is necessary and the timing of IPR within the treatment. If the IPR procedure is conducted in a stage of the treatment that is too early or too late, the treatment professional may have poor access to the surfaces of the one or more teeth intended to be removed. Further, the treatment professional may inaccurately remove material from the surface of the tooth resulting in an undesired tooth shape, a tooth surface that does not fit properly against another tooth, and potentially having to perform additional IPR procedures and/or other procedures to fix the overlap or newly created underlap.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A illustrates an example of a digital dental model according to one or more embodiments of the present disclosure.

FIG. **1**B illustrates an example of a target digital dental model according to one or more embodiments of the present disclosure.

FIG. **1**C illustrates an example of a revised target digital dental model according to one or more embodiments of the present disclosure.

FIGS. **2**A-**2**B illustrate examples of a ridge endpoint reference line on each of a first tooth and a second tooth according to one or more embodiments of the present disclosure.

FIG. **2**C illustrates an example of a y-axis reference line on each of a first tooth and a second tooth according to one or more embodiments of the present disclosure.

FIG. **3** is a flow chart illustrating an example of a process for IPR planning according to one or more embodiments of the present disclosure.

FIG. **4** illustrates an example of a method for IPR planning according to one or more embodiments of the present disclosure.

FIG. **5** illustrates an example of a system for IPR planning according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

Embodiments of the present disclosure include computing device related, system, and method embodiments for interproximal reduction planning. For example, one or more embodiments include a computing device implemented method that includes identifying a first tooth and a second tooth in a digital dental model with an overlap in a target position in a treatment plan, identifying a reference line on each of the first tooth and the second tooth, revising the treatment plan wherein a position of each of the first tooth and the second tooth is determined in which the reference line of the first tooth and the reference line of the second tooth are aligned and the first tooth and the second tooth are not overlapping, and prescribing interproximal reduction based on the determined position of each of the first tooth and the second tooth.

In the following detailed description of the present disclosure, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration how a number of embodiments of the disclosure may be practiced. These embodiments are described in sufficient detail to enable those of ordinary skill in the art to practice a number of embodiments of this disclosure, and it is to be understood that other embodiments may be utilized and that changes may be made without departing from the scope of the present disclosure.

As will be appreciated, elements shown in the various embodiments herein can be added, exchanged, and/or eliminated so as to provide a number of additional embodiments of the present disclosure. In addition, as will be appreciated, the proportion and the relative scale of the elements provided in the figures are intended to illustrate the embodiments of the present disclosure, and should not be taken in a limiting sense. As used herein, "a", "at least one", "a number of" something can refer to one or more such things.

Although the overarching term "orthodontics" is used herein, the present disclosure may relate to treatments of an orthognathic nature. For example, in cases including treatment of a patient's underlying skeletal structure, teeth may be rearranged by surgically repositioning underlying bones that hold the teeth in order to achieve a desired final bite arrangement. In both orthodontic and orthognathic treatment approaches, alignment of the teeth may be evaluated pre-, mid-, and/or post-treatment.

Treatment professionals, such as a clinician, typically select a treatment plan for a patient's teeth based upon experience with certain types of physical features and/or appliances. In some embodiments, a dental treatment plan can include orthodontic treatment planning functions and/or appliances. IPR procedures are often left to be done at the end of a dental treatment, however, there may not be enough space to properly align the teeth at this point in the treatment.

IPR planning within a dental treatment plan can be beneficial, for example, in that planning for IPR can consider the ease of performing IPR. The planning of IPR can result in more accurate access to the IPR regions as compared to non-planning for IPR, more accurate removal of the IPR regions as compared to non-planning for IPR, and the elimination of additional procedures to fix errors caused by inaccurate IPR timing, among other benefits.

In various embodiments, with the use of computer device executable instructions, a treatment professional can establish a treatment plan having a target position for a number of teeth of a particular patient. With this target position in mind, a first tooth and a second tooth needing IPR can be identified and IPR can be virtually planned at a point during the process of moving the teeth to the target position that is desirable for performing IPR.

Digital dental models from a scan of a patient's dentition can be provided with computer-aided design and/or manufacturing systems, including tooth-treatment systems. A digital dental model representing an initial tooth arrangement may be obtained in a variety of ways.

For example, the patient's teeth may be imaged to obtain digital data using direct and/or indirect structured light, X-rays, three-dimensional X-rays, lasers, destructive scanning, computer-aided tomographic images or data sets, magnetic resonance images, intra-oral scanning technology, photographic reconstruction, and/or other imaging techniques. The digital dental model can include an entire mouth tooth arrangement, some, but not all teeth in the mouth, and/or it can include a single tooth.

A positive model and/or negative impression of the patient's teeth or a tooth may be scanned using an X-ray, laser scanner, destructive scanner, structured light, and/or other range acquisition system to produce the initial digital dental model. The data set produced by the range acquisition system may be converted to other formats to be compatible with the software which is used for manipulating images within the data set, as described herein.

Referring now to FIG. **1**A, there is illustrated an example of a digital dental model **100** according to one or more embodiments of the present disclosure. As described herein, the digital dental model **100** can be obtained prior to treatment or at an intermediate state of treatment (e.g., before treatment has been completed). The digital dental model **100** can include an initial position of a first tooth **101** and a second tooth **102** or a number of teeth.

FIG. **1**B illustrates a target digital dental model **104** according to one or more embodiments of the present disclosure. The target digital dental model **104** can be created by modifying the digital dental model **100** according to one or more treatment goals. The target digital dental model **104** can include a target position in a treatment plan of a first tooth **101** and a second tooth **102** and a target position of a number of teeth, for example. The one or more treatment goals can be case-specific (e.g., specific to the particular patient on which the digital dental model **100** was based).

In some embodiments, a first tooth **101** and a second tooth **102** can be identified in the digital dental model **100** with an overlap **103** in a target position in a treatment plan. An overlap **103** can include an overlap with a neighboring tooth **102** along an arch of the target digital dental model **104** and a potential collision in the target position with a neighboring tooth **102**, among many others. An overlap **103** that can be remedied through IPR can be approximately 0.50 millimeters, for example. For instance, in such an example, approximately 0.25 millimeters of the first tooth **101** and 0.25 millimeters of the second tooth **102** can be removed via an IPR procedure. In some IPR procedures, the amount of tooth material removed can be different for two adjacent teeth (e.g., 0.25 mm and 0.15 mm).

FIG. **1**C illustrates a revised target digital dental model **106** after IPR prescription according to one or more embodiments of the present disclosure. The revised target digital dental model **106** can be created by modifying the digital dental model **100** and the target digital dental model **104** according to the IPR prescribed at the determined position.

For example, the revised target digital dental model **106** can include a digital removal of the portion of the first tooth **101** and second tooth **102** overlapping **103** in the target digital dental model **104** and identified to be removed at the determined position. In some embodiments, the removed portion **103** can include a total of approximately 0.25 millimeters of the first tooth **101** and 0.25 millimeters of the second tooth **102** for a total removed portion **103** of approximately 0.50 millimeters, for example. Such an amount of removal can be beneficial for the cosmetic appearance and/or the structural integrity of the teeth **101**,**102** prescribed for IPR.

In various embodiments of the present disclosure, the treatment plan can include a plurality of stages. For example, at a determined position of the revised treatment plan a number of additional stages can be inserted into the revised treatment plan.

A first additional stage can include removing an overlap of the first tooth **101** and the second tooth **102**. For example, the first additional stage can include expanding the arch of the first tooth **101** and the second tooth **102**. The expansion can include digitally changing a mesial position or distal position of at least one tooth of the first tooth **101** and the second tooth **102**, among others. A second additional stage can include aligning the first tooth **101** and the second tooth **102** for IPR.

In various embodiments, the process can include reclining the first tooth **101** and the second tooth **102** to a revised target position in the revised target digital model **106** after IPR prescription. In various embodiments, a number of teeth without overlap can be reclined to a revised target position in the revised target digital dental model **106** after IPR prescription.

In some embodiments, the target digital dental model **104** and revised target digital dental model **106** can reflect an intermediate tooth movement within a treatment plan. Such a tooth movement may be useful during a particular process within the treatment plan (e.g., IPR, extraction, etc.).

In some embodiments, the digital dental model **100**, target digital dental model **104**, and the revised target digital dental model **106** can be displayed via a user interface in three dimensions.

FIGS. **2**A-**2**C illustrate examples of a reference line on each of a first tooth and a second tooth. In various embodiments of the present disclosure, a reference line on each of the first tooth and the second tooth can be identified.

FIGS. **2**A-**2**B illustrate examples of a ridge endpoint reference line **216-1**, **216-2**, **218-1**, **218-2** on each of a first tooth **208** and a second tooth **210** according to one or more embodiments of the present disclosure. The first tooth **208** and the second tooth **210** can be among a number of teeth **208**, **210**, **212**, and **214** in a treatment plan.

In one or more embodiments, a reference line can be identified on each of the first tooth **208** and the second tooth **210** by calculating the distance difference between ridge endpoints of each of the first tooth **208** and the second tooth **210** at the determined position **209** of each tooth and at the target position **211** of each tooth. For example, the distance **216-1** between the ridge endpoints of the first tooth **208** and the second tooth **210** at the determined position **209** and the distance **216-2** at the target position **211** can be measured. The distance **216-1** at the determined position **209** minus the distance **216-2** at the target position **211** can then be calculated.

In various embodiments of the present disclosure, the reference lines of the first tooth **208** and the second tooth **210** may be aligned if the distance difference between the ridge endpoints at the determined position **209** and the target position **211** is within a threshold difference. The threshold distance difference can be a distance that is predetermined (e.g., by a treatment professional). For example, a threshold distance difference can be a distance of 0.75 millimeters.

Such a threshold may indicate that the teeth are aligned vertically and/or laterally such that IPR can be performed. For instance, a distance difference threshold of 0.75 millimeters may result in the identification and alignment of teeth that may be severely misaligned for IPR if IPR is left to be performed at the end of the dental treatment. Teeth severely misaligned for IPR can include teeth that would be difficult or impossible for a treatment professional to perform an IPR procedure on at the end of the dental treatment.

FIG. **2**C illustrates an example **219** of a y-axis reference line **228**, **226** on each of a first tooth **220** and a second tooth **224** according to one or more embodiments of the present disclosure. In one or more embodiments, a reference line **228**, **226** can be identified on each of a first tooth **220** and a second tooth **224** by calculating the y-axis of the first tooth **228** and the y-axis of the second tooth **226**. For example, the y-axis of the first tooth **228** and the second tooth **226** can include the mesial-distal local axis of a tooth.

In various embodiments of the present disclosure, the y-axis reference lines **228**, **226** of the first tooth **220** and the second tooth **224** may be aligned if the angle **230** between the y-axis of the first tooth **228** and the y-axis of the second tooth **226** is within a threshold angle β. The threshold angle β can be an angle that is predetermined (e.g., by a treatment professional). For example, the threshold angle θ may be 20 degrees. Such a threshold may indicate that the teeth are aligned vertically and/or laterally such that IPR can be performed.

In various embodiments of the present disclosure, the reference lines of the first tooth **220** and the second tooth **224** may be aligned if the angle **230** between the y-axis of the first tooth **228** and the y-axis of the second tooth **226** at a target position is greater than the angle **230** of the y-axis of the first tooth **228** and the y-axis of the second tooth **226** at the determined position. For example, if the angle **230** at the target position is approximately 30 degrees and the angle **230** at the determined position is approximately 25 degrees, then the first tooth **220** and the second tooth **226** may be aligned for IPR at the determined position. A greater y-axis angle at the target position may indicate that the teeth are aligned vertically and/or laterally such that IPR can be performed.

In some embodiments of the present disclosure, the reference lines of the first tooth and second tooth may be aligned if the calculated difference between the ridge endpoints of the first tooth and the second tooth at the determined position and the target position is within a threshold distance, if the angle between the first tooth and the second tooth at the determined position is within a threshold angle, or if the calculated angle between the y-axis of the first tooth and the second tooth at the target position is greater than the calculated angle at the determined position, or any combination thereof. The combination may indicate that the teeth are aligned vertically and/or laterally such that IPR can be performed. For example, the combination may result in more accurate identification and alignment of teeth that may be severely misaligned for an IPR procedure if IPR is left to be performed at the end of the dental treatment.

FIG. **3** is a flow chart illustrating an example of a process **332** for IPR planning according to one or more embodiments of the present disclosure. At **334**, a digital dental model can be created from data received. For example, a digital dental model can be created from a scan of a patient's dentition and provided with computer-aided design and/or manufacturing systems, including tooth-treatment systems.

The digital dental model can include data for a number of teeth. The digital dental model can include an initial digital dental model or an intermediate position of a digital dental model, for example.

At **336**, a treatment plan can be computed from the digital dental model. The treatment plan can include moving a number of teeth to a target position of the treatment plan. In various embodiments of the present disclosure, the treatment plan can include a plurality of stages.

At **338**, a determination can be made as to whether a first tooth overlaps with a second tooth in the target position. A first tooth overlapping with a second tooth can include a potential collision in the target position with a neighboring tooth. In various embodiments of the present disclosure, the first tooth that overlaps with the second tooth can be overlapping with a number of teeth on different sides of the first tooth.

In response to no identified overlapping teeth in the target position, at **348**, the process **332** can identify that IPR planning may not be required.

In response to determining that a first tooth overlaps with a second tooth, at **340**, a determination can be made as to whether the first tooth and the second tooth are misaligned for IPR in the initial position. Misalignment in the initial position can include no access for a treatment professional to perform IPR. For example, the distance between the IPR region of the first tooth and the IPR region of the second tooth may be too small for a treatment professional to perform IPR.

In response to a determination that the teeth are not misaligned for IPR in the initial position, at **346**, IPR can be prescribed in the initial position. In response to a determination that the teeth are misaligned for IPR in the initial position, the treatment plan can be revised. At **342**, a stage can be identified or a stage can be added in the treatment plan wherein the reference line of the first tooth and the reference line of the second tooth are aligned and the teeth are not colliding. At **344**, IPR can be prescribed on the first tooth and the second tooth at the identified or added stage.

FIG. **4** illustrates an example of a method **492** for IPR planning according to one or more embodiments of the present disclosure. At **494**, a first tooth and a second tooth with an overlap in a target position of a treatment plan can be identified in a digital dental model. For example, an overlap can include a calculated overlap or a potential collision in a target position with a neighboring tooth.

In various embodiments of the present disclosure, an overlap between a first tooth and a second tooth can be disregarded wherein the first tooth and the second tooth are at least one of a molar and/or a premolar. Disregarding molars and/or premolars can include not aligning the identified overlapping molars and/or premolars for IPR and not planning IPR for the overlapping molars and/or premolars. For example, an overlapping molar and/or premolar can include an overlap between a first molar and a second molar, an overlap between a first molar and a first premolar, and an overlap between a first premolar and a second premolar.

At **496**, a reference line can be identified for each of the first tooth and the second tooth. In various embodiments, a reference line can include a distance between the ridge endpoints of each of the first tooth and the second tooth. In various embodiments, a reference line can include a y-axis of each of the first tooth and the second tooth. A y-axis of each tooth can include a mesial-distal local axis of the tooth, for example.

A reference line between the ridge endpoints of each of a first tooth and a second tooth can be used on teeth with well defined ridges in various embodiments of the present disclosure. For example, a ridge endpoint reference line may be determined on incisor and/or canine teeth, and a ridge endpoint reference line may not be suitable on molar and/or premolar teeth.

At **498**, a position of each of the first tooth and the second tooth is determined in which the reference line of the first tooth and the reference line of the second tooth are aligned and not overlapping.

At **499**, IPR can be prescribed based on the determined position of each of the first tooth and the second tooth. For example, IPR prescription can include indicating in a treatment plan that an IPR procedure is to be done at a particular stage of the plan based on this analysis.

FIG. **5** illustrates an example of a system **550** for IPR planning according to one or more embodiments of the present disclosure. In the system **550** in FIG. **5**, the system **550** includes a computing device **552** having a number of components coupled thereto. The computing device **552** includes a processor **554** and a memory **556**. The memory **556** can have various types of information including data **558** and executable instructions **560**, as discussed herein.

The processor **554** can execute instructions **560** that are stored on an internal or external non-transitory computer device readable medium (CRM). A non-transitory CRM, as used herein, can include volatile and/or non-volatile memory. Volatile memory can include memory that depends upon power to store information, such as various types of dynamic random access memory (DRAM), among others. Non-volatile memory can include memory that does not depend upon power to store information.

Memory **556** and/or the processor **554** may be located on the computing device **552** or off the computing device **552**, in some embodiments. As such, as illustrated in the embodiment of FIG. **5**, a system **550** can include a network interface **562**. Such an interface **562** can allow for processing on another networked computing device, can be used to obtain information about the patient, and/or can be used to obtain data and/or executable instructions for use with various embodiments provided herein.

As illustrated in the embodiment of FIG. **5**, a system **550** can include one or more input and/or output interfaces **580**. Such interfaces **580** can be used to connect the computing device **552** with one or more input and/or output devices **582**, **584**, **586**, **588**, and **590**.

For example, in the embodiments illustrated in FIG. **5**, the system **550** can include connectivity to a scanning device **582**, a camera dock **584**, an input device **586** (e.g. a mouse, a keyboard, etc.), a display device **588** (e.g., a monitor), a printer **590**, and/or one or more other input devices. The input/output interface **580** can receive executable instructions and/or data, storable in the data storage device (e.g., memory **556**), representing a digital dental model, target digital dental model, and/or a revised target digital dental model of a patient's dentition.

In some embodiments, the scanning device **582** can be configured to scan one or more physical molds of a patient's dentition. In one or more embodiments, the scanning device **582** can be configured to scan the patient's dentition directly. The scanning device **582** can be configured to input data into the computing device **552**.

In some embodiments, the camera dock **584** can receive an input from an imaging device (e.g., a two-dimensional or three-dimensional imaging device) such as a digital camera, a printed photograph scanner, or other suitable imaging device. The input from the imaging device can, for example, be stored in memory **556**.

The processor **554** can execute instructions to provide a visual indication of a treatment plan and/or provided stage for IPR planning on the display **588**. The computing device **550** can be configured to allow a treatment professional or other user to input treatment goals. Input received can be sent to the processor **554** as data and/or can be stored in memory **556**.

Such connectivity can allow for the input and/or output of data and/or instructions among other types of information. Although some embodiments may be distributed among various computing devices within one or more networks, such systems as illustrated in FIG. **5**, can be beneficial in allowing for the capture, calculation, and/or analysis of information discussed herein.

The processor **554**, in association with the data storage device (e.g., memory **556**), can be associated with data **558**. The processor **454**, in association with the memory **456**, can store and/or utilize data **558** and/or execute instructions **560** for IPR planning. Such data **558** can include the digital dental model, target digital dental model, and/or the revised target digital dental model.

The processor **554** coupled to the memory **556** can analyze a target digital dental model of a treatment plan, including tooth data for a number of teeth, to identify if a first tooth in the number of teeth overlaps with a neighboring tooth along an arch of the target digital dental model. The overlap can include a collision of a first tooth and a second tooth in a target position within a target digital dental model, for example.

The processor **554** coupled to the memory **556** can revise the treatment plan to remove the overlap and align the first tooth and neighboring tooth for IPR, prescribe IPR on the first tooth and the neighboring tooth in the aligned position, and recline the first tooth and the neighboring tooth to a revised target position in a revised target digital dental model after IPR prescription. Removing the overlap can include expanding the arch by digitally changing a mesial position or a distal position of at least one tooth of the first tooth and the neighboring tooth, among many others. Prescription of IPR can include a notification of the aligned position and a digital representation of the region of the first tooth and the second tooth removed by the IPR procedure prescribed, for example.

In various embodiments of the present disclosure, the processor **554** coupled to the memory **556** can recline a number of teeth without undesired overlap in the digital dental model to the revised target position in the revised target digital dental model after IPR prescription.

Although specific embodiments have been illustrated and described herein, those of ordinary skill in the art will appreciate that any arrangement calculated to achieve the same techniques can be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments of the disclosure.

It is to be understood that the above description has been made in an illustrative fashion, and not a restrictive one. Combination of the above embodiments, and other embodiments not specifically described herein will be apparent to those of skill in the art upon reviewing the above description.

The scope of the various embodiments of the disclosure includes any other applications in which the above structures and methods are used. Therefore, the scope of various embodiments of the disclosure should be determined with reference to the appended claims, along with the full range of equivalents to which such claims are entitled.

In the foregoing Detailed Description, various features are grouped together in example embodiments illustrated in the figures for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the embodiments of the disclosure require more features than are expressly recited in each claim.

Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

What is claimed is:

1. A method of interproximal reduction (IPR) treatment planning for a patient's dentition, the method comprising:

identifying an overlap between a first tooth and an adjacent second tooth in the patient's dentition at a target position of a treatment plan, wherein the treatment plan includes a plurality of stages for incrementally moving the patient's dentition, and wherein the target position corresponds to a desired position of the first tooth and the second tooth in accordance with the treatment plan;

identifying a first distance along a ridge endpoint reference lines between a ridge endpoint of the first tooth and a ridge endpoint of the second tooth at a provided position of the treatment plan, wherein the provided position corresponds to a position of the first tooth and the second tooth prior to treatment or at an intermediate stage of the treatment plan;

calculating, using a processor, a distance difference between the first distance and a second distance, wherein the second distance is along the ridge endpoint reference line between the ridge endpoint of the first tooth and the ridge endpoint of the second tooth at the target position;

determining that the distance difference is outside of a threshold difference indicating that the first tooth and the second tooth are misaligned for IPR in the provided position;

revising, using the processor, the treatment plan by adding one or more additional stages to the plurality of stages of the treatment plan, the one or more additional stages characterized in that the distance difference between the first and second distances is within the threshold difference;

prescribing IPR on one or both of the first tooth and the second tooth at the one or more additional stages of the treatment plan; and

outputting the revised treatment plan.

2. The method of claim **1**, wherein identifying the overlap comprises disregarding a molar or a premolar overlap between a first molar and a second molar or between a premolar and a molar.

3. The method of claim **1**, wherein the one or more additional stages of the treatment plan includes expanding an arch of the first tooth and the second tooth.

4. The method of claim **3**, wherein expanding the arch includes digitally changing a mesial position or distal position of one or both of the first tooth and the second tooth.

5. The method of claim **1**, wherein revising the treatment plan further comprises adding a second additional stage that includes aligning the first tooth and the second tooth for IPR.

6. The method of claim **1**, wherein revising the treatment plan comprises removing the overlap by changing a mesial position of one or both of the first tooth and the second tooth.

7. The method of claim **1**, wherein revising the treatment plan comprises removing the overlap by changing a distal position of one or both of the first tooth and the second tooth.

8. The method of claim **1**, wherein revising the treatment plan comprises reclining one or both of the first tooth and the second tooth to a revised target position.

9. The method of claim **1**, wherein determining that the distance difference is outside of the threshold difference comprises determining that the first tooth and the second tooth are misaligned vertically, laterally, or vertically and laterally such that IPR cannot be performed.

10. A system for interproximal reduction (IPR) treatment planning, the system comprising:

a processor; and

a memory coupled to the processor and configured to direct the processor to:

identify an overlap between a first tooth and a second tooth in a patient's dentition at a target position of a treatment plan, wherein the treatment plan includes a plurality of stages for incrementally moving the patient's dentition, and wherein the target position corresponds to a desired position of the first tooth and the second tooth in accordance with the treatment plan;

identify a first distance along a ridge endpoint reference lines between a ridge endpoint of the first tooth and a ridge endpoint of the second tooth at a provided position of the treatment plan, wherein the provided position corresponds to a position of the first tooth and the second tooth prior to treatment or at an intermediate stage of the treatment plan;

calculate, using the processor, a distance difference between the first distance and a second distance, wherein the second distance is along the ridge endpoint reference line between the ridge endpoint of the first tooth and the ridge endpoint of the second tooth at the target position;

determine that the distance difference is outside of a threshold difference indicating that the first tooth and the second tooth are misaligned for IPR in the provided position;

revise, using the processor, the treatment plan by adding one or more additional stages to the plurality of stages of the treatment plan, the one or more additional stages characterized in that the distance difference between the first and second distances is within the threshold difference;

prescribe IPR on one or both of the first tooth and the second tooth at the one or more additional stages of the treatment plan;

output the revised treatment plan.

11. The system of claim **10**, wherein the one or more additional stages of the treatment plan includes expanding an arch of the first tooth and the second tooth.

12. The system of claim **11**, wherein expanding the arch includes digitally changing a mesial position or distal position of one or both of the first tooth and the second tooth.

13. The system of claim **10**, wherein determining that the distance difference is outside of the threshold difference comprises determining that the first tooth and the second tooth are misaligned vertically, laterally, or vertically and laterally such that IPR cannot be performed.

14. The system of claim **10**, wherein revising the treatment plan further comprises adding a second additional stage that includes aligning the first tooth and the second tooth for IPR.

15. The system of claim **10**, wherein revising the treatment plan comprises removing the overlap by changing a mesial position of one or both of the first tooth and the second tooth.

16. The system of claim **10**, wherein revising the treatment plan comprises removing the overlap by changing a distal position of one or both of the first tooth and the second tooth.

17. The system of claim **10**, wherein revising the treatment plan comprises reclining one or both of the first tooth and the second tooth to a revised target position.

18. A method of interproximal reduction (IPR) treatment planning for a patient's dentition, the method comprising:

identifying an overlap between a first tooth and a second tooth in the patient's dentition at a target position of a treatment plan, wherein the treatment plan includes a plurality of stages for incrementally moving the patient's dentition, and wherein the target position corresponds to a desired position of the first tooth and the second tooth in accordance with the treatment plan;

identifying a first distance along a ridge endpoint reference lines between a ridge endpoint of the first tooth and a ridge endpoint of the second tooth at a provided position of the treatment plan, wherein the provided position corresponds to a position of the first tooth and the second tooth prior to treatment or at an intermediate stage of the treatment plan;

calculating, using a processor, a distance difference between the first distance and a second distance, wherein the second distance is along the ridge endpoint reference line between the ridge endpoint of the first tooth and the ridge endpoint of the second tooth at the target position, wherein the ridge endpoint reference line includes vertical and lateral dimensions;

determining that the distance difference is outside of a threshold difference indicating that the first tooth and the second tooth are misaligned for IPR in the provided position;

revising, using the processor, the treatment plan by adding one or more additional stages to the plurality of stages of the treatment plan, the one or more additional stages characterized in that the distance difference between the first and second distance is within the threshold difference;

prescribing IPR on one or both of the first tooth and the second tooth at the one or more additional stages of the treatment plan; and

outputting the revised treatment plan with the prescribed IPR.

19. The method of claim **1**, wherein the ridge endpoint reference line includes vertical and lateral dimensions.

20. The method of claim **1**, further comprising:

determining that the second tooth and a third tooth, adjacent to the second tooth, is misaligned for IPR based on a second distance difference between ridge endpoints of the second and third tooth being outside the threshold difference; and

revising the treatment plan such that the second distance difference is within the threshold difference.

* * * * *